

ISOGENY, ZETA-EQUIVALENCE, AND THE PICARD TORSOR IN FUNCTION FIELD BOST–CONNES SYSTEMS

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ABSTRACT. Paper XVIII of the series showed that the partition function of the function field Bost–Connes system is the Weil zeta function, and asked whether the system can distinguish two curves with the same zeta function but non-isomorphic Jacobian groups. The answer is yes, and we locate precisely which layer of the system does it.

The proposed exact sequence $1 \rightarrow \text{Jac}(C)(\mathbb{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$ is not the right one: $\text{Jac}(C)(\mathbb{F}_q) = \text{Pic}^0(C)$ is a *quotient* of the compact part $\mathbb{I}_K^0/K^\times$, not a subgroup of $G_{\mathcal{P}_K}$, and the correct picture is a two-step filtration with the much larger $\prod_P \mathcal{O}_P^\times/\overline{\mathbb{F}_q^\times}$ at the bottom. What is true, and is what the disambiguation needs, is the identity

$$G_{\mathcal{P}_K}/\overline{U_S} \cong \widehat{\text{Pic}(C)} \cong \widehat{\mathbb{Z}} \times \text{Jac}(C)(\mathbb{F}_q), \quad U_S = \prod_P \mathcal{O}_P^\times,$$

coming from $\mathbb{I}_K/(K^\times U) = \text{Div}(C)/\text{Prin}(C) = \text{Pic}(C)$ together with F. K. Schmidt’s theorem that C has a divisor of degree 1. Since $\widehat{\mathbb{Z}}$ is torsion free,

$$\text{Jac}(C)(\mathbb{F}_q) = (G_{\mathcal{P}_K}/\overline{U_S})_{\text{tors}},$$

so the Jacobian group is recovered from the *pair* $(G_{\mathcal{P}_K}, U_S)$ — data the groupoid supplies but the partition function does not.

Hence the disambiguation theorem holds, at the Cartan-pair level: a gauge-equivariant isomorphism of systems forces $\text{Jac}(C_1)(\mathbb{F}_q) \cong \text{Jac}(C_2)(\mathbb{F}_q)$, while the partition function forces only $Z_{C_1} = Z_{C_2}$, that is, isogeny. The separation is not vacuous. By brute-force enumeration over $\mathbb{F}_p$ for $p \leq 23$ we exhibit many isogeny classes containing several group structures; for $p = 13$ and trace -2 there are three, $\mathbb{Z}/16$, $\mathbb{Z}/2 \times \mathbb{Z}/8$ and $\mathbb{Z}/4 \times \mathbb{Z}/4$, all with the same zeta function.

Two cautions. Over a function field $S = \mathcal{P}_K$ is massively *not* norm-separated — $N(D) = q^{\deg D}$ identifies all divisors of equal degree — so Theorem 2.5 of Paper XIII does not apply at all and the gauge-equivariant hypothesis of Theorem 4.2 of Paper XIV is not a convenience but a necessity. And the proposed third tier, recovery of C itself by Torelli, does not follow from anything here: what the system supplies is abelian class field theory data, and the abelianized Galois group is known not to determine a global field in the number field case. We record it as open, not proved.

1. THE CORRECT FILTRATION

Let $C/\mathbb{F}_q$ be smooth projective geometrically irreducible, $K = \mathbb{F}_q(C)$, $S = \mathcal{P}_K$ the set of closed points. The framework is that of [5] transported to function fields as in Paper XVII of the series, including its §1 convention that $G_{\mathcal{P}_K}$ denotes the profinite completion in the degree direction; Papers XIII, XIV, XVII and XVIII, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place. Function fields have no archimedean places, so $\mathbb{I}_S = \mathbb{I}_K$, $K_S^\times = K^\times$ and

$$G_{\mathcal{P}_K} = \mathbb{I}_K/\overline{K^\times}, \quad U_S = \prod_P \mathcal{O}_P^\times.$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XIX-isogeny/](#).

Proposition 1.1 (Two-step filtration). *There are exact sequences of topological groups*

$$(1.1) \quad 1 \longrightarrow \prod_P \mathcal{O}_P^\times / \overline{\mathbb{F}_q^\times} \longrightarrow \mathbb{I}_K^0 / K^\times \longrightarrow \mathrm{Pic}^0(C)(\mathbb{F}_q) \longrightarrow 1,$$

$$(1.2) \quad 1 \longrightarrow \mathbb{I}_K^0 / K^\times \longrightarrow G_{\mathcal{P}_K} \xrightarrow{\deg} \widehat{\mathbb{Z}} \longrightarrow 1,$$

where $\mathbb{I}_K^0$ denotes the degree-zero ideles. The group $\mathbb{I}_K^0 / K^\times$ is compact.

Proof. Standard class field theory for function fields: (1.2) is the degree map, whose image after profinite completion is $\widehat{\mathbb{Z}}$; (1.1) is the quotient of the degree-zero idele class group by the unit ideles, whose cokernel is the degree-zero divisor class group. $\square$

Remark 1.2 (The proposed sequence). The sequence $1 \rightarrow \mathrm{Jac}(C)(\mathbb{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$ is not correct. By (1.1), $\mathrm{Jac}(C)(\mathbb{F}_q) = \mathrm{Pic}^0(C)$ is a *quotient* of the compact part, not a subgroup, and the kernel in (1.2) is $\mathbb{I}_K^0 / K^\times$, an extension of $\mathrm{Pic}^0(C)$ by the enormous group $\prod_P \mathcal{O}_P^\times / \overline{\mathbb{F}_q^\times}$. The Jacobian group is a subquotient of $G_{\mathcal{P}_K}$, and exhibiting it requires naming U_S .

2. THE JACOBIAN GROUP AS A TORSION SUBGROUP

Theorem 2.1 (Recovery from the pair). *With $U_S = \prod_P \mathcal{O}_P^\times \leq G_{\mathcal{P}_K}$,*

$$(2.1) \quad G_{\mathcal{P}_K} / \overline{U_S} \cong \widehat{\mathrm{Pic}(C)} \cong \widehat{\mathbb{Z}} \times \mathrm{Jac}(C)(\mathbb{F}_q),$$

and consequently

$$\mathrm{Jac}(C)(\mathbb{F}_q) \cong \left(G_{\mathcal{P}_K} / \overline{U_S} \right)_{\mathrm{tors}}.$$

Proof. $\mathbb{I}_K / (K^\times U_S)$ is by definition $\mathrm{Div}(C) / \mathrm{Prin}(C) = \mathrm{Pic}(C)$, and passing to profinite completions gives the first isomorphism in (2.1). By F. K. Schmidt's theorem [3] C has a divisor of degree 1, so the degree map $\mathrm{Pic}(C) \rightarrow \mathbb{Z}$ splits and $\mathrm{Pic}(C) \cong \mathbb{Z} \oplus \mathrm{Pic}^0(C)$ with $\mathrm{Pic}^0(C) = \mathrm{Jac}(C)(\mathbb{F}_q)$ finite; completing gives $\widehat{\mathbb{Z}} \times \mathrm{Jac}(C)(\mathbb{F}_q)$. Finally $\widehat{\mathbb{Z}}$ is torsion free, so the torsion subgroup of the product is the finite factor. $\square$

Corollary 2.2 (What each layer sees). *The partition function determines $Z_C(T)$ (Theorem 4.1 of Paper XVIII), hence $|\mathrm{Jac}(C)(\mathbb{F}_q)| = P(1)$ but only its order. The pair $(G_{\mathcal{P}_K}, U_S)$ determines $\mathrm{Jac}(C)(\mathbb{F}_q)$ as an abstract finite abelian group, by Theorem 2.1.*

3. DISAMBIGUATION

Theorem 3.1 (Isogeny disambiguation). *Let C_1, C_2 be curves over $\mathbb{F}_q$ with $Z_{C_1} = Z_{C_2}$ but $\mathrm{Jac}(C_1)(\mathbb{F}_q) \not\cong \mathrm{Jac}(C_2)(\mathbb{F}_q)$. Then:*

- (1) *the two systems have the same partition function, so no invariant built from the KMS temperature dependence alone distinguishes them;*
- (2) *nevertheless the systems are not isomorphic gauge-equivariantly: there is no isomorphism $\mathcal{A}_{C_1, \mathcal{P}_{K_1}} \rightarrow \mathcal{A}_{C_2, \mathcal{P}_{K_2}}$ intertwining the full gauge actions in the sense of Paper XIV (its Definition 4.1) — the actions of the duals of the grading groups, of which the circle action of Paper XVIII and the flow σ are quotients.*

Proof. (1) is Theorem 4.1 of Paper XVIII. For (2), suppose Φ is such an isomorphism. The argument of Theorem 4.2 of Paper XIV — the integer-valued gauge cocycle and its bisections — uses only the étale groupoid picture and the freeness of the grading monoid, and applies verbatim over a function field; it gives $\Phi(C(Y_{S_1})) = C(Y_{S_2})$ together with a monoid isomorphism

$\theta : J_{S_1} \rightarrow J_{S_2}$, i.e. a θ -equivariant homeomorphism $h : Y_{S_1} \rightarrow Y_{S_2}$ of the semigroup dynamical systems. Now consider the *deep stratum*

$$Y_\infty := \bigcap_{0 \neq D \in J_S} D \cdot Y_S = \{[x, g] : x = 0\} \cong G_{\mathcal{P}_K} / \overline{U_S},$$

canonically defined from the dynamics, on which $D \in J_S$ acts by the translation $[0, g] \mapsto [0, \sigma_D g]$: the deep stratum is a torsor over the compactified Picard group $G_{\mathcal{P}_K} / \overline{U_S} \cong \widehat{\text{Pic}(C)}$ of Theorem 2.1, and the translations $\{\sigma_D\}$ run through the effective classes, which generate $\text{Pic}(C)$ and are dense in the completion. The closure of the translation monoid in $\text{Homeo}(Y_\infty)$ is therefore a compact group acting simply transitively — a closed subsemigroup of a compact group is a subgroup — and it is isomorphic, as a topological group, to $G_{\mathcal{P}_K} / \overline{U_S}$. Since h maps $Y_{\infty,1}$ onto $Y_{\infty,2}$ intertwining the translation monoids, it intertwines their closures, giving a topological group isomorphism $G_{\mathcal{P}_{K_1}} / \overline{U_{S_1}} \cong G_{\mathcal{P}_{K_2}} / \overline{U_{S_2}}$. Taking torsion subgroups, Theorem 2.1 forces $\text{Jac}(C_1)(\mathbb{F}_q) \cong \text{Jac}(C_2)(\mathbb{F}_q)$, contrary to hypothesis. (Note that the homeomorphism type of Y_∞ alone would see nothing: all these spaces are Cantor sets, and indeed $|\text{Jac}|$ is already matched by (1); it is the translation structure that carries the group.) $\square$

Remark 3.2 (Gauge-equivariance is necessary here). Over a function field $N(D) = q^{\deg D}$, so N identifies all divisors of the same degree and $S = \mathcal{P}_K$ is as far from norm-separated as possible: the relation lattice $L(S)$ of Paper XIV (its Definition 1.2) has infinite rank. Consequently Theorem 1.4 of Paper XIII, which identifies $C(Y_S)$ with $\mathcal{A}^\sigma$ under norm-separation, is unavailable, and $\mathcal{A}^\sigma$ is a large non-commutative algebra — as is the fixed-point algebra of the circle action, which coincides with it. Theorem 3.1 therefore rests on Theorem 4.2 of Paper XIV and its *full* gauge-equivariant hypothesis; equivariance for the circle action alone is the same as σ -equivariance and does not feed that theorem. In this setting the hypothesis is not a technical convenience but the only route available.

Example 3.3 (Explicit isogeny classes with several group structures). Enumerating all $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$ and grouping by trace t — equal t being equivalent to equal zeta function, hence to isogeny [4] — gives, for instance:

p	t	group structures of $E(\mathbb{F}_p)$
11	0	$\mathbb{Z}/12, \mathbb{Z}/2 \times \mathbb{Z}/6$
13	-2	$\mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8, \mathbb{Z}/4 \times \mathbb{Z}/4$
13	5	$\mathbb{Z}/9, \mathbb{Z}/3 \times \mathbb{Z}/3$
17	2	$\mathbb{Z}/16, \mathbb{Z}/2 \times \mathbb{Z}/8, \mathbb{Z}/4 \times \mathbb{Z}/4$
19	-7	$\mathbb{Z}/27, \mathbb{Z}/3 \times \mathbb{Z}/9$
23	-8	$\mathbb{Z}/32, \mathbb{Z}/2 \times \mathbb{Z}/16$

Each row is a single isogeny class, so a single partition function, containing curves whose systems are pairwise non-isomorphic by Theorem 3.1. The phenomenon is abundant, not exceptional: it occurred for most traces at every prime $11 \leq p \leq 23$ examined.

4. THE HIERARCHY, WITH THE THIRD TIER OPEN

Theorem 4.1 (Two tiers). *For the function field system $(\mathcal{A}_{C, \mathcal{P}_K}, \mathbb{T}, \sigma)$:*

- **Tier 1, thermodynamic.** *The partition function recovers $Z_C(T)$, hence the Weil polynomial, the genus, all point counts $\#C(\mathbb{F}_{q^n})$, and $|\text{Jac}(C)(\mathbb{F}_q)|$. Two systems have the same partition function iff the curves are zeta-equivalent, that is iff their Jacobians are isogenous over $\mathbb{F}_q$ [4].*

- **Tier 2, Cartan pair.** A gauge-equivariant isomorphism recovers the diagonal with its semigroup action, hence the deep stratum with its translation structure, hence the compactified Picard group and $\text{Jac}(C)(\mathbb{F}_q)$ as an abstract group, strictly refining Tier 1 by Example 3.3.

Remark 4.2 (The third tier is not established). The proposal’s third tier — recovery of C itself, “the Torelli geometric isomorphism” — does not follow from what is proved here, and we do not believe it follows from abelian data at all. What the system supplies is the abelianized Galois group with its Frobenius classes: class field theory data. In the number field case the abelianized absolute Galois group is known not to determine the field — Onabe, and Angelakis–Stevenhagen [1], exhibit infinitely many imaginary quadratic fields sharing the same $\text{Gal}(K^{ab}/K)$ — so no purely abelian invariant can give a Torelli statement. Our data is richer than the bare group, carrying the Frobenius classes indexed by S together with their degrees; whether that suffices is exactly the question addressed in the literature on characterizing global fields by their L -series, which we have not been able to consult here. We state Tier 3 as open and make no claim.

Remark 4.3 (What Tier 2 does and does not give). Theorem 2.1 recovers $\text{Jac}(C)(\mathbb{F}_q)$ as an abstract finite abelian group. It does not recover the principal polarization, which is what Torelli’s theorem requires (for the arithmetic of pairings and polarizations over finite fields see [2]), nor the Galois module structure of the ℓ -adic Tate module beyond its characteristic polynomial. So even a complete solution at Tier 2 would leave a gap before Tier 3, and the gap is exactly the polarization.

5. ASSESSMENT

Statement	Status	Reference
$1 \rightarrow \text{Jac}(\mathbb{F}_q) \rightarrow G_{\mathcal{P}_K} \rightarrow \widehat{\mathbb{Z}} \rightarrow 1$	false: wrong kernel	Rem. 1.2
two-step filtration	true	Prop. 1.1
$G_{\mathcal{P}_K}/\overline{U_S} \cong \widehat{\mathbb{Z}} \times \text{Jac}(\mathbb{F}_q)$	true	Thm. 2.1
$\text{Jac}(\mathbb{F}_q) = (G_{\mathcal{P}_K}/\overline{U_S})_{\text{tors}}$	true	Thm. 2.1
partition function $\iff$ isogeny	true	Thm. 4.1
system distinguishes isogenous curves	yes, at the Cartan pair	Thm. 3.1
$S = \mathcal{P}_K$ norm-separated	no; $L(S)$ of infinite rank	Rem. 3.2
gauge-equivariance needed	yes, necessarily	Rem. 3.2
Tier 3: $C_1 \cong C_2$	open; not from abelian data	Rem. 4.2

The question left open in §5 of Paper XVIII is answered: zeta-equivalence is *not* the equivalence realized by these systems. The partition function sees exactly the isogeny class, and the Cartan pair sees strictly more, namely the isomorphism class of $\text{Jac}(C)(\mathbb{F}_q)$ as a group. The mechanism is the identity $\mathbb{I}_K/(K^\times U) = \text{Pic}(C)$, which is class field theory rather than operator algebra; the operator-algebraic content is only that a gauge-equivariant isomorphism recovers the diagonal with its semigroup action, which is Theorem 4.2 of Paper XIV — the passage from there to the Picard group is the translation structure of the deep stratum.

We are left with a clean picture of the layers and an honest gap at the top. Thermodynamics gives the isogeny class; the Cartan pair gives the Jacobian group; and the curve itself is out of reach of both, because the polarization is nowhere in the data and because abelian class field theory is known to be insufficient for reconstructing global fields.

Three questions. Does the pair (G_S, U_S) with the Frobenius classes determine more than $\text{Jac}(C)(\mathbb{F}_q)$ — for instance the full group $\text{Pic}(C)$ with its effective cone, which is closer to geometry? Second, Theorem 2.1 used F. K. Schmidt to split the degree map; is the splitting canonical

enough that the isomorphism $\text{Jac}(C)(\mathbb{F}_q) \cong (\cdot)_{\text{tors}}$ is functorial, or does the choice of degree-one divisor matter for the disambiguation? Third, and this is the natural next target, is there a curve pair with the same zeta *and* the same $\text{Jac}(\mathbb{F}_q)$ but non-isomorphic curves — for which Tier 2 would also fail, and the question of Tier 3 would become sharp rather than speculative?

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